

Article

Identification of Different Ecomorphotypes of *Coilia nasus* in the Dawanzhou Section of the Yangtze River

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Abstract: X-ray electron probe microanalyzer technology was used to study the microchemistry and habitat history of *Coilia nasus* collected from the Dawanzhou section of the Yangtze River between May and June 2023. The Sr/Ca ratio from the otolith core to the otolith diameter was low (640–1100 μm), representing the first stage of development. In the second stage, *C. nasus* exhibited two distinct types. The first type, which included individuals 5HK05 and 6HK03, exclusively inhabited brackish estuarine waters. The second type, comprising 13 individuals, resided in higher-salinity seawater environments ($\text{Sr/Ca} > 7$). Furthermore, individuals 5HK01, 5HK03, 5HK07, and 6HK05 displayed a phase with a high Sr/Ca ratio compared to other fish. Freshwater coefficient analysis indicated that *C. nasus* in the Dawanzhou water area was unlikely to continue upstream to Dongting Lake in the middle reaches of the Yangtze River but may have entered Poyang Lake through its mouth or reproduced in its upper reaches. Analysis of sexual maturity and migration history suggested that the Dawanzhou section primarily serves as a migration channel for *C. nasus*, with a potential spawning ground for this high-value fish located nearby.

Keywords: *Coilia nasus*; otolith microchemistry; freshwater coefficient; Dawanzhou section of Yangtze River



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Key Contribution: In this study, the life history of *Coilia nasus* in the Dawanzhou section of the Yangtze River was investigated using otolith microchemistry. The results indicated that *C. nasus* in this area are likely to enter Poyang Lake for spawning or continue reproducing along the Yangtze River to the spawning grounds in the upper reaches of Poyang Lake but are unlikely to return to the waters of Dongting Lake.

1. Introduction

The estuarine tapertail anchovy (*Coilia nasus*; Temminck and Schlegel, 1846) belongs to the order Clupeiformes and the family Engraulidae. *C. nasus* is protein-rich, nutritionally balanced, and has an appealing taste with various health benefits, making it a highly valuable fish species. The annual breeding peak of *C. nasus* occurs between May and August [1]. As an anadromous fish, adult *C. nasus* migrate up the Yangtze River from the estuary and enter coastal lakes for breeding. However, due to habitat deterioration and overfishing, *C. nasus* can no longer breed in the waters of Poyang and Dongting Lakes [2]. The coexistence of various ecological types further complicates effective research and conservation efforts.

Otoliths record the life history and habitat transitions of fish, forming during the hatching period of fertilized eggs and subsequently growing around the core sediment [3]. Previous studies have shown that otolith strontium (Sr) serves as a useful marker for mapping fish migration history, as otolith Sr levels and Sr:Ca ratios are generally positively correlated with habitat salinity in freshwater, estuarine, and marine fish [4,5]. Sr:Ca ratios are typically high in seawater but generally low in freshwater environments [6–8]. Using microprobe analysis of otolith Sr and Sr:Ca ratio levels, researchers have delineated variable patterns of habitat utilization in fish species, particularly anadromous fish [9].

To better assess and effectively protect *C. nasus* populations in the Yangtze River, Jiang et al. [10] classified *C. nasus* into five ecomorphotype resource groups using otolith microchemistry: anadromous migratory and freshwater-resident long-supermaxilla *C. nasus*, anadromous migratory and freshwater-resident short-supermaxilla *C. nasus*, and land-locked long-supermaxilla *C. nasus taihuensis*. Therefore, modern approaches such as otolith microchemistry provide a more objective means of understanding habitat history, migration patterns, and ecomorphotype composition of *C. nasus* compared to traditional morphological studies.

The lower reaches of the Yangtze River refer to the 940 km section of the main channel extending from the mouth of Poyang Lake to the East China Sea. The Dawanzhou section of the Yangtze River, located in Jiangxi Province, is adjacent to the mouth of Poyang Lake and serves as the only route for *C. nasus* to migrate between the Yangtze River estuary and Poyang Lake. As the sole inlet to Poyang Lake and the primary migration corridor for *C. nasus* within the lake, the Hukou water area plays a crucial role in maintaining and replenishing *C. nasus* populations. Therefore, studying the *C. nasus* population in Poyang Lake and its surrounding waters is of great ecological significance. Additionally, this area functions as a key migration channel for *C. nasus* from the lower to the middle reaches of the Yangtze River.

To date, research on *C. nasus* in the Yangtze River has primarily focused on the estuarine and lower reaches [11–13]. However, the migration history and population dynamics of *C. nasus* in this critical habitat remain largely unknown. A comprehensive analysis of the migration patterns and ecomorphotypes of *C. nasus* in the Dawanzhou section is necessary to gain a deeper understanding of its movement and population trends. Such insights can provide essential baseline data for conserving anchovy resources, identifying spawning grounds, establishing germplasm or nature reserves, and formulating targeted conservation policies. This need has become particularly urgent following the implementation of China's 10-year fishing ban along the Yangtze River.

Otolith microchemical technology offers distinct advantages in analyzing the temporal and spatial aspects of fish life history, including habitat history and migration patterns [14–17]. Therefore, the present study aimed to utilize this technology to examine the microchemical characteristics of *C. nasus* in the Dawanzhou River segment during May and June, thereby supplementing fundamental data on anadromous *C. nasus*.

Furthermore, studies on the impact of the 10-year fishing ban on *C. nasus* remain limited. Ma et al. [18] found that following the implementation of the fishing ban, the *C. nasus* population in the Yangtze River estuary has doubled annually. Similarly, Jiang et al.'s survey of *C. nasus* spawning grounds in Poyang Lake revealed that the average proportion of *C. nasus* in catches after the fishing ban (2019–2020) was approximately 43 times higher than before the ban (2014–2018). The increase in *C. nasus* resources also raises the possibility of upstream migration toward Dongting Lake, an aspect that warrants further investigation.

This study sought to assess the conservation effects of the 10-year fishing ban on *C. nasus*, a “flagship species” that enhances public awareness of ecosystem protection. Additionally, by reconstructing the habitat history and ecological role of *C. nasus* in the

unique environment of the Dawanzhou section post-fishing ban, this research provides a recent case study for evaluating the responses of migratory fish populations in the river–lake confluence zones of the Yangtze River. The findings are expected to offer valuable technical support for policymakers in developing more effective fish conservation strategies during the ongoing fishing ban.

2. Materials and Methods

2.1. Materials

The Dawanzhou section of the Yangtze River forms a “Y” shape and is located at the junction of Poyang Lake and the Yangtze River. Fifteen *C. nasus* samples were collected from the Dawanzhou water area in Jiangzhou Town, Chaisang District, Jiujiang City, near the mouth of Poyang Lake (Figure 1). All samples belonged to the traditional long-supermaxilla *C. nasus* type, with 10 collected in May 2023 and 5 in June 2023.

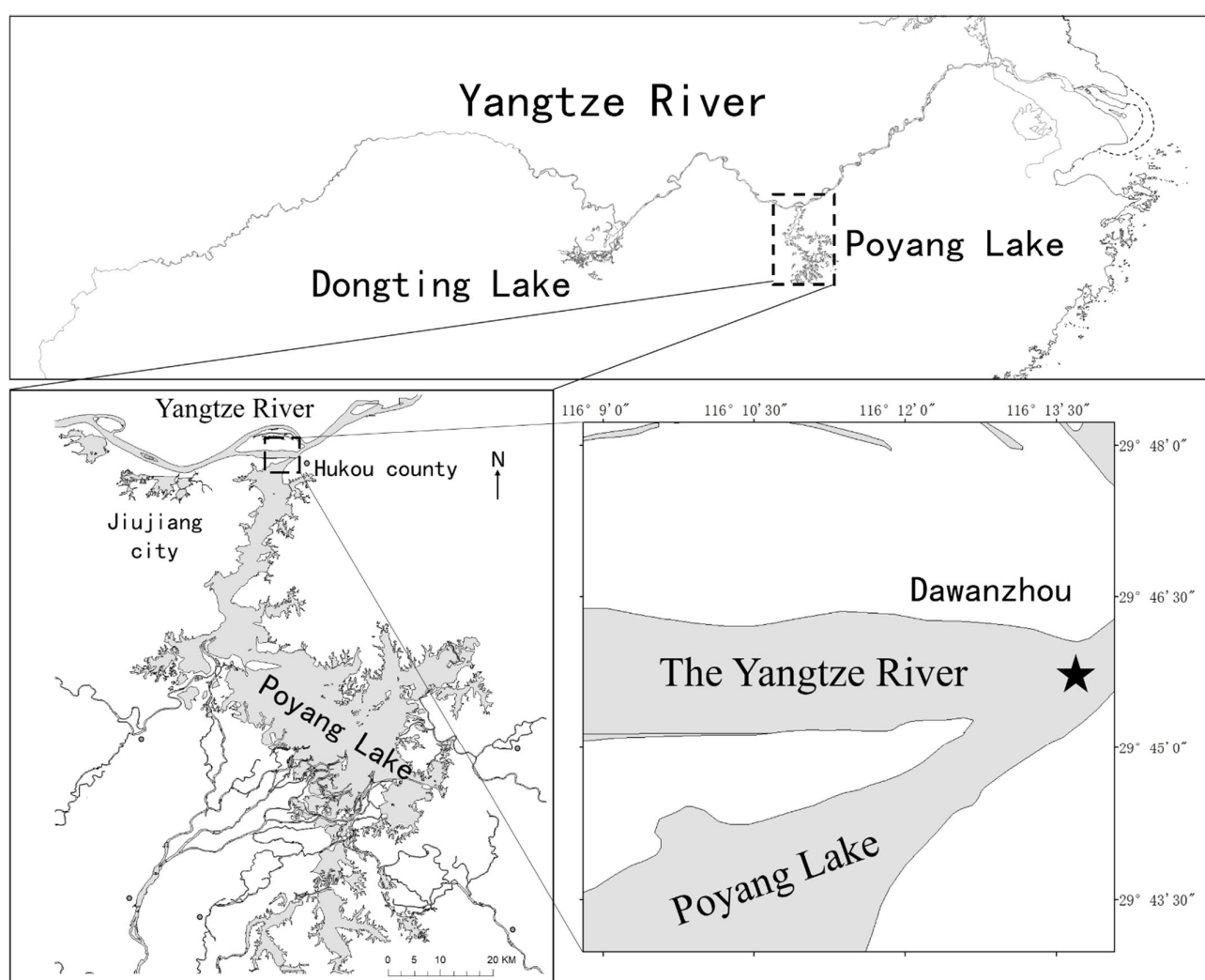


Figure 1. *Coilia nasus* sampling sites (★) in the Dawanzhou section of the Yangtze River. The dashed line indicates the approximate area of the Yangtze Estuarine brackish water regime, based on the study of Zhao et al. [19].

After capture, the samples were refrigerated and stored at -20°C for subsequent processing. Before dissection, biological data were recorded, including fork length, body weight, and the supermaxilla-to-head length ratio (Table 1). The maturity of *Coilia nasus* was determined according to Li et al. [2]. The left sagittal otolith was extracted for microchemical analysis.

Table 1. Biological information of *Coilia nasus* in the Dawanzhou section of the Yangtze River.

Sample Code	Fork Length (mm)	Body Weight (g)	Jaw Length/Head Length	Sex/Maturity	Age (Years) *
5HK01	316.43	106.03	1.19	♂/III	3
5HK02	327.30	112.67	1.11	♀/III	3
5HK03	344.94	162.07	1.26	♀/III	3
5HK04	275.11	95.81	1.11	♂/III	3
5HK05	313.83	97.73	1.23	♀/III	3
5HK06	311.45	95.88	1.13	♂/III	3
5HK07	342.04	130.34	1.14	♀/III	3
5HK08	345.52	131.35	1.13	♀/III	3
5HK09	377.92	97.30	1.12	♂/II	3
5HK10	345.50	115.48	1.18	♀/III	3
6HK01	342.40	130.00	1.14	♀/III	3
6HK02	318.40	114.50	1.10	♀/IV	3
6HK03	275.30	52.00	1.21	♂/III	3
6HK04	283.10	79.50	1.13	♀/III	3
6HK05	275.60	73.50	1.12	♀/IV	3

* Age based on the otolith annulus.

2.2. Otolith Microchemical Analysis

The otolith pre-treatment process was as follows: First, the otolith was embedded in a mold using epoxy resin (Epofix, Struers, Copenhagen, Denmark). After curing, excess resin above the otolith was removed with a mill (Discoplan-TS model, Struers, Denmark), followed by grinding with a 70 µm emery wheel until the otolith was exposed. Further grinding with a 35 µm emery wheel was performed until the otolith core was exposed. Once the core was exposed, the otolith was polished using a polishing machine (LaboPol, Struers, Denmark) with an organic woven disc until its surface was smooth. The polished otolith was then placed in a Milli-Q water ultrasonic bath for 5 min.

After cleaning and drying, the dried otolith flakes were coated with a carbon film (36 A, 25 s) using a vacuum-coating machine (JEE-420, JEOL Co., Ltd., Tokyo, Japan). Following pre-treatment, an X-ray electron probe microanalyzer (EPMA) (JXA-8100, JEOL Co., Ltd., Tokyo, Japan) was used to perform a quantitative line analysis of the Sr and Ca contents along a straight line from the core to the edge of the otolith along its longest diameter. Referring to the method of Yang et al. [20], the EPMA parameters were set as follows: acceleration voltage of 15 kV, electron beam current of 20 nA, beam diameter of 5 µm, and a 15 s dwell time at each point. Dot determination was carried out continuously with a 20 µm spacing. Strontium titanate (SrTiO₃) and calcium carbonate (CaCO₃) were used as standard samples. The results of the quantitative line analysis were expressed as the Sr–Ca ratio (Sr/Ca × 1000).

Based on EPMA technology, Yang et al. [20] studied *Coilia ectenes* in Taihu Lake, as well as *Coilia mystus* and *Coilia nasus* in the Yangtze River estuary, and established criteria for identifying Sr/Ca values and Sr content surface distribution maps for different salinity habitats. The Sr/Ca values corresponding to freshwater, brackish estuarine, and seawater habitats were <3, 3–7, and >7, respectively. The colors in the Sr content surface distribution map corresponded to blue, green, yellow, and red.

Following EPMA analysis, all otolith samples were re-polished to remove the carbon film from the surface. A 5% EDTA solution was then used to etch the otoliths for 10 s, after which the otolith age was determined once the rings became clear. The otolith age count was obtained from a study by Li et al. [21].

2.3. Statistical Analysis

Data were summarized and mapped using Microsoft Excel 2016. To more intuitively observe the trend of the Sr/Ca ratio in *C. nasus*, the method described by Stirnimann [22] was applied, using Sequential *t*-test Analysis of Regime Shifts to perform trend transformations on the results of the quantitative line analysis. The truncation length was set to 5, the Huber weight to 1, and the confidence level (P) to 0.1.

To assess the significance of differences in Sr/Ca values between different *C. nasus* life-history stages, SPSS26.0 was used to perform non-parametric tests (Mann–Whitney U-tests). The freshwater coefficient (F_C), established by Jiang et al. [23] in a study of *C. nasus* populations in different coastal waters of China, was introduced to reflect the dependence of *C. nasus* on freshwater habitats in the Dawanzhou section of the Yangtze River. The freshwater coefficient is defined as $F_C = L_f/L_T$, where L_f is the length of the first freshwater stage, as indicated by the low Sr/Ca ratio line in the otolith microchemical line analysis or the low-Sr bluish central regions in the Sr mapping analysis, and L_T is the radius of the entire otolith microchemical line analysis measurement line along the longest axis of each otolith from the core.

3. Results

Based on otolith annulus readings, all 15 fish in this study were three-year-old individuals. The specific changes in Sr/Ca values are shown in Table 2.

Table 2. Variation of otolith Sr/Ca ratio of *Coilia nasus* from the Dawanzhou section of the Yangtze River.

Sampling Code	Sr/Ca Ratios Variation	Distance from Otolith Core (μm)	Number of Detected Points	Sr: Ca × 1000 Mean (±SD)	F_C
5HK01	1	0–940	48	1.54 ± 0.67	0.48
	2	960–1160	11	4.82 ± 1.29	
	3	1180–1320	8	8.01 ± 0.87	
	4	1340–1960	32	5.98 ± 1.22	
	5	1980–2000	2	2.53 ± 0.69	
5HK02	1	0–700	36	1.39 ± 0.69	0.33
	2	720–2120	71	5.51 ± 1.45	
5HK03	1	0–1040	53	1.40 ± 0.67	0.45
	2	1060–1480	22	4.89 ± 1.37	
	3	1500–1600	6	7.99 ± 0.77	
	4	1620–2300	35	4.85 ± 0.91	
5HK04	1	0–1080	55	1.61 ± 0.54	0.54
	2	1100–2000	46	4.39 ± 1.07	
5HK05	1	0–1040	53	1.81 ± 0.83	0.49
	2	1060–2160	56	3.52 ± 0.88	
	3	2160–2180	1	0.70 ± 0	
5HK06	1	0–1040	53	1.40 ± 0.56	0.55
	2	1060–1860	40	5.26 ± 1.57	
5HK07	1	0–1100	56	1.46 ± 0.57	0.49
	2	1120–1600	25	4.75 ± 0.83	
	3	1620–1880	14	7.61 ± 1.22	
	4	1900–2240	18	5.26 ± 0.86	
5HK08	1	0–980	50	1.54 ± 0.55	0.39
	2	1000–2500	76	4.49 ± 1.22	

Table 2. Cont.

Sampling Code	Sr/Ca Ratios Variation	Distance from Otolith Core (μm)	Number of Detected Points	Sr: Ca $\times 1000$ Mean ($\pm\text{SD}$)	F_c
5HK09	1	0–920	47	1.41 ± 0.70	0.41
	2	940–2240	66	4.81 ± 0.95	
5HK10	1	0–1020	52	1.62 ± 0.63	0.47
	2	1040–2180	58	5.52 ± 0.89	
6HK01	1	0–640	33	1.53 ± 0.50	0.44
	2	660–1820	59	4.82 ± 0.94	
	3	1840–2160	17	1.46 ± 0.92	
6HK02	1	0–700	36	1.43 ± 0.56	0.45
	2	720–1700	50	4.06 ± 1.23	
	3	1720–1840	7	1.80 ± 0.69	
6HK03	1	0–1000	51	1.58 ± 0.57	0.56
	2	1020–1740	37	4.32 ± 0.90	
	3	1760–1980	12	2.91 ± 1.22	
	4	2000–2140	8	4.29 ± 1.00	
6HK04	1	0–1000	51	1.46 ± 0.54	0.49
	2	1020–2040	52	4.98 ± 1.56	
6HK05	1	0–640	33	1.69 ± 0.49	0.49
	2	660–820	9	5.57 ± 1.40	
	3	840–1100	14	7.41 ± 0.60	
	4	1120–1700	30	5.46 ± 1.05	
	5	1700–1720	1	2.84 ± 0	

According to the results of the line transect analysis of Sr/Ca values from the core to the edge (Figure 2), the Sr/Ca values of all *C. nasus* remained consistently low ($1.39 [\pm 0.69]$ – $1.81 [\pm 0.83]$) from the otolith core to a distance of 640–1100 μm from the core, representing the first stage. In the subsequent stages, the Sr/Ca values of *C. nasus* in May and June exceeded 3.0. However, due to differences in fluctuation patterns, the fish could be categorized into two types.

The first type included individuals 5HK05 and 6HK03. In the second stage, their Sr/Ca values began to increase, gradually exceeding 3.0 ($3.52 [\pm 0.08]$ and $4.32 [\pm 0.90]$), and remained stable.

The second type consisted of 13 individuals (5HK01–5HK04, 5HK01–5HK10, 6HK01, 6HK02, 6HK04, and 6HK05), exhibiting more complex Sr/Ca fluctuations. In addition to the low values observed in the first stage, these individuals also had high-value stages with Sr/Ca values exceeding 3.0 and, in some cases, even surpassing 7.0. Among these, 5HK01, 5HK03, 5HK07, and 6HK05 could be further categorized into a subgroup characterized by a short-lived high Sr/Ca stage ($7.41 [\pm 0.60]$ – $8.01 [\pm 0.87]$).

Subsequently, the Sr/Ca values of some individuals (5HK01, 5HK04, 5HK06, 5HK08, 5HK09, and 6HK01–6HK05) began to decline and eventually fell below 3.0. In contrast, other individuals (5HK02, 5HK03, 5HK07, and 5HK10) exhibited more complex patterns, where Sr/Ca values decreased toward the otolith edge but never fell below 3.0.

All *C. nasus* in this study exhibited life history stages with Sr/Ca values exceeding 3.0, indicating a history of river–sea migration. Thus, they were classified as anadromous migratory long-supermaxilla *C. nasus*. Moreover, the first annulus of *C. nasus* formed after entering the brackish waters of estuaries during the second stage, while the second annulus developed before returning to freshwater habitats. The freshwater coefficient (F_c) values of the otolith samples captured in May and June were 0.46 and 0.49, respectively.

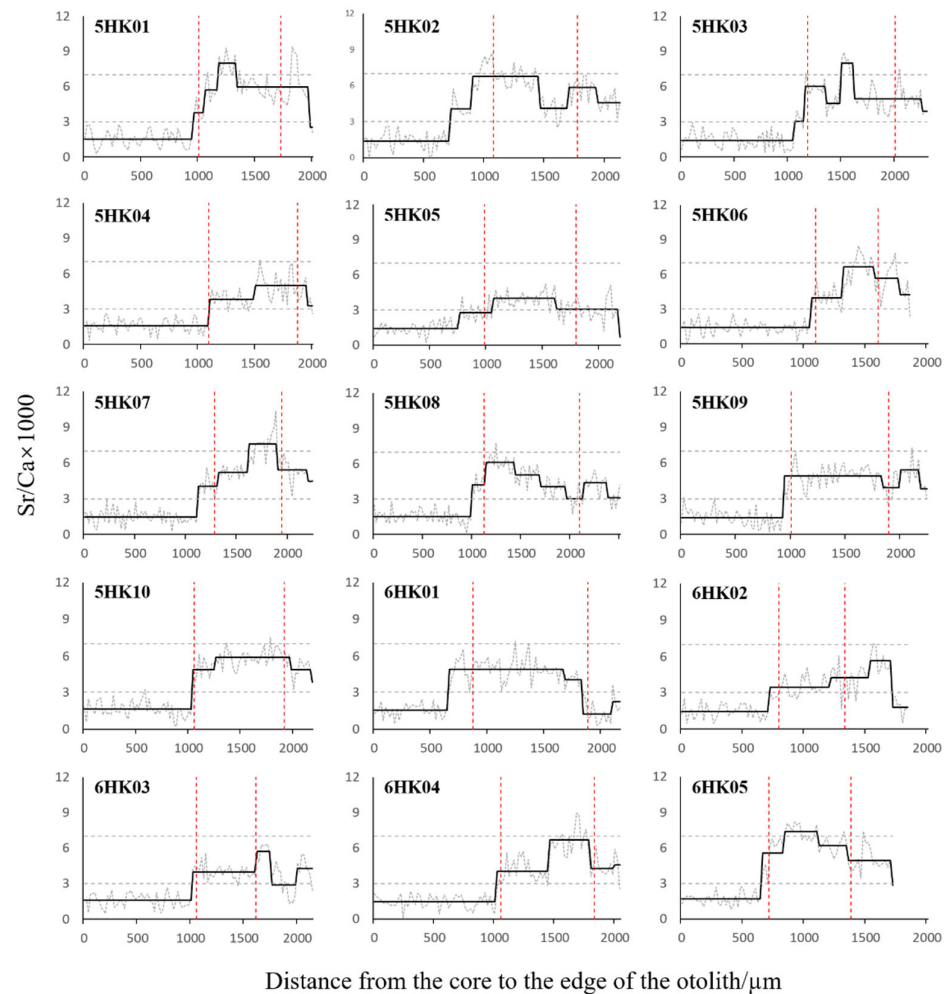


Figure 2. Sr/Ca values of *Coilia nasus* in the Dawanzhou section of the Yangtze River from core to edge line transect analysis. Gray lines represent the Sr/Ca value fluctuation, black lines represent the trend conversion result, and red dotted lines represent *C. nasus* age rings.

4. Discussion

C. nasus is a migratory fish species in the Yangtze River and can serve as a “flagship species” for evaluating the effectiveness of aquatic life protection measures in the river. It should also be considered one of the key beneficial species in the context of the fishing ban. Numerous successful studies have utilized otolith microchemical techniques to reconstruct fish migration histories and life history patterns [24–26]. These studies provide a solid foundation for accurately tracing the habitat history of *C. nasus* in the Dawanzhou section of the Yangtze River. In this study, the sampled *C. nasus* were identified as long-supermaxilla *C. nasus* (supermaxilla: head length > 1). Based on the line transect analysis of otolith from the core to the edge, the Sr/Ca values of most individuals exhibited a sudden increase from a low-value stage (<3) to a high-value stage (3–7 or >7), followed by a gradual decline back to a low value. This pattern indicated that the hatching and early development of these individuals occurred in freshwater, and then they subsequently migrated to the estuary’s brackish water or high-salinity seawater for fattening and growth and eventually returned upstream to the Dawanzhou section of the Yangtze River, where they were captured. These findings confirm their typical anadromous migration behavior.

However, different individuals entered high Sr/Ca ratio areas at different times. For example, strain 5HK04 entered the high Sr/Ca area during the formation of its first annual annulus, whereas strain 5HK02 entered this area much later. This variation may be because

C. nasus migrated to the estuarine fragmentation zone for growth and development during its early and juvenile stages, where it was influenced by the brackish water in the estuary [27].

Additionally, although samples 5HK02, 5HK03, 5HK07, and 5HK10 were also captured in the Dawanzhou section of the Yangtze River, line transect analysis revealed that their Sr/Ca values did not fall below 3. This suggests that the tracing time for these *C. nasus* individuals was relatively short and that a “lag effect” may have occurred in Sr and Ca deposition in otoliths [28].

Based on these findings, we speculate that the *C. nasus* population captured in May consisted of individuals that migrated back in batches, whereas the population captured in June likely comprised individuals that migrated back simultaneously. The Sr/Ca values of the otolith margins decreased to 1.49–2.84. Furthermore, all *C. nasus* samples captured in this study formed their first annual annulus after entering the second developmental stage or after a period of fattening in brackish water habitats in estuaries.

This variation in the timing of migration from freshwater to high-salinity environments highlights the diversity of early life history patterns among migratory *C. nasus*. Similar diversity in migration timing has been observed in many other migratory fish species [29,30].

In this study, the water area of the Dawanzhou section of the Yangtze River was located at the junction of Poyang Lake and the Yangtze River, near the mouth of Poyang Lake, and exhibited a “Y”-shaped fork. Changes in the fish community structure in this aquatic area influence the fishery function of Poyang Lake and the ecosystem’s self-regulation capacity. Therefore, understanding the migration and life history patterns of *C. nasus* in this region is crucial.

Poyang Lake is the largest freshwater lake in China and a typical “huff-and-puff” lake, serving as an important feeding and spawning ground for anadromous fish. It is also a key habitat for other economically significant and endangered fish species along the Yangtze River, where fishing has been banned for ten years. Each year, during spring and summer, mature *C. nasus* migrate from the coast to lakes along the middle and lower reaches of the Yangtze River to reproduce [10]. According to a previous study, the spawning period of *C. nasus* near the mouth of Poyang Lake occurs around June [31], which aligns with the sampling time in this study.

As shown in Table 1, except 5HK09, which exhibited sexual maturity in May in the Dawanzhou section of the Yangtze River, all other individuals were at Stage III maturity. In contrast, the sexual maturity of 6HK02 and 6HK05, captured in June, had reached Stage IV. According to Li et al. [2], when *C. nasus* reaches Stage IV sexual maturity, the synthesis and accumulation of nutrients, such as yolk, are complete. If ecological conditions are suitable for spawning, *C. nasus* undergoes spawning and fertilization [32], suggesting that spawning grounds may exist near the sampled waters.

Additionally, the *C. nasus* population captured in May had not yet reached sexual maturity, meaning they were not yet ready for spawning. Given that the primary purpose of anadromous fish migration is reproduction [33] and that long-distance anadromous species exhibit ecological behaviors enabling them to return precisely to their natal spawning grounds [34], it is likely that the Dawanzhou section of the Yangtze River serves as a migration corridor for the *C. nasus* population collected in May.

F_C was established by Jiang et al. [23] while studying *C. nasus* populations in different coastal waters of China. This coefficient primarily reflects the degree of dependence of migratory fish on freshwater habitats and the characteristics of the *C. nasus* population. It can also serve as an indicator of *C. nasus* origin to some extent.

In the present study, the F_C of *C. nasus* captured in May was $0.46 (\pm 0.07)$, while that of individuals captured in June was $0.49 (\pm 0.05)$. Due to the unique topography of the Dawanzhou section of the Yangtze River, *C. nasus* can not only enter Poyang Lake but also

continue swimming along the Yangtze River, reaching the upper reaches of Poyang Lake or Dongting Lake for breeding and spawning.

According to a previous analysis, the gonads of *Coilia anchovies* captured in May, as well as some captured in June, had not yet reached maturity, indicating the possibility of further migration. When Xuan [35] studied *C. nasus* in Dongting Lake, the F_C values of two individuals were 0.60 and 0.56, respectively, which were significantly different from those observed in the present study. Furthermore, F_C can also provide insights into the distance of parent fish from the estuary. Therefore, it is unlikely that the *C. nasus* population in the Dawanzhou section, as observed in this study, will continue migrating upstream to Dongting Lake for breeding.

However, Jiang et al. [23] found that one group of *C. nasus* in coastal China had an F_C value of 0.48, which closely matched the F_C of the *C. nasus* population captured in June in the present study. This similarity suggests that both groups most likely originated from the same spawning ground in Poyang Lake.

Notably, our previous analyses revealed variations in the Sr/Ca ratios of freshwater samples from different sections of the Yangtze River catchment. For example, the Sr/Ca values were 5.27 ± 1.19 in Poyang Lake, 8.45 ± 0.30 in the middle reaches of the Yangtze River, 6.12 ± 1.14 in the lower reaches, and 7.93 ± 1.54 in Dongting Lake. However, all these values were significantly lower than the Sr/Ca ratio of estuarine water (19.00 ± 6.82).

Future studies should focus on investigating and identifying spawning grounds around the Dawanzhou section of the Yangtze River and the connected areas of Poyang Lake further. This research would provide a scientific basis for establishing a potential *C. nasus* conservation area at this critical river-lake confluence along the Yangtze River.

All the *C. nasus* samples in this study were typical anadromous migratory long-supermaxilla *C. nasus*. The Sr/Ca values at the otolith core (<3) indicated that they all hatched in freshwater spawning fields; however, their life history strategies differed. Based on these differences, the life history of all *C. nasus* samples in this study could be categorized into two types (Figure 3).

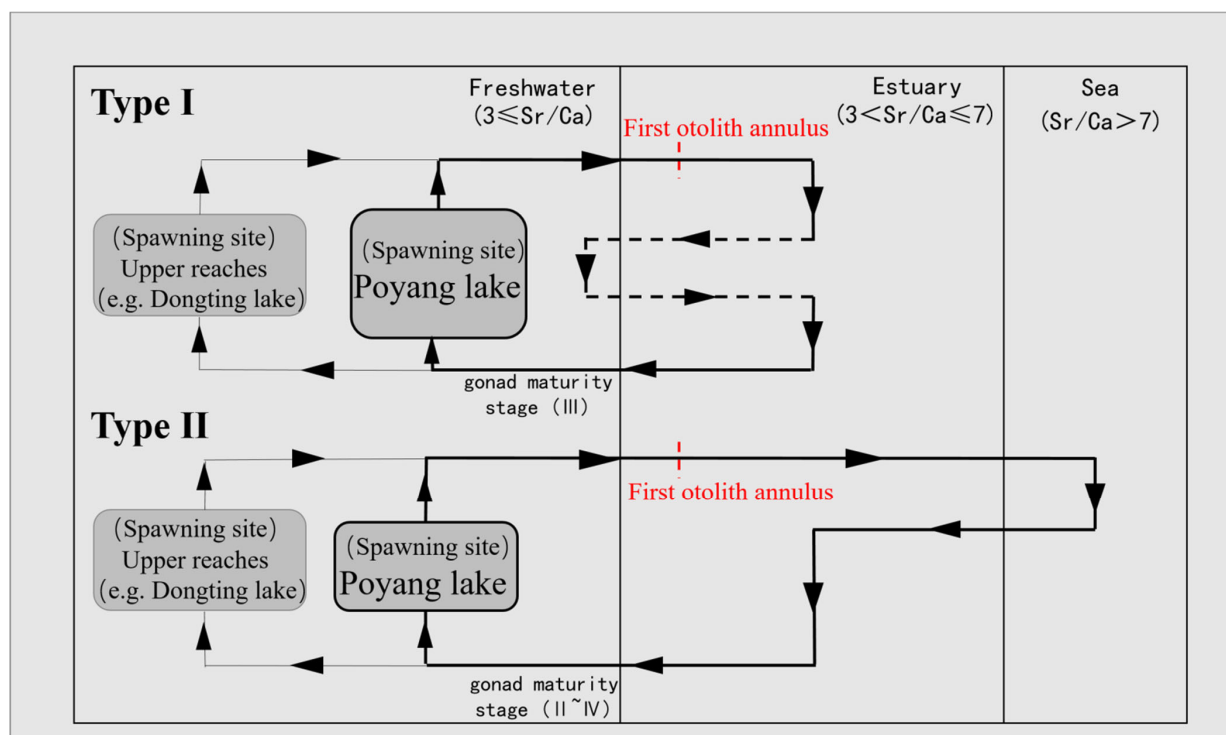


Figure 3. Migration model of *Coilia nasus* in the Dawanzhou section of the Yangtze River. The lines with arrows represent the possible dispersion patterns during the non-spawning migratory phases.

Samples 5HK05 and 6HK03 belonged to the first type. These individuals did not enter the sea after reaching the brackish water of the estuary for growth but instead continued to migrate back and forth between freshwater and brackish water. The first annulus formed shortly after they entered the brackish water of the estuary. These fish were captured at the sampling site while returning to their spawning grounds, at which time their gonads were only at Stage III, indicating that they had not yet reached maturity or the spawning stage [36]. The distribution of individuals with advanced gonadal development can be used to identify spawning grounds. Moreover, individuals with lower gonadal maturity are less likely to expend energy on migration before reaching sexual maturity. Therefore, *C. nasus* individuals in the first type may have originated from Poyang Lake or spawning grounds in the upper reaches of Poyang Lake.

In contrast to the first type, *C. nasus* in the second life-history model entered seawater habitats for development but did not remain in seawater for an extended period. Additionally, the gonads of individuals 6HK02 and 6HK05 had reached Stage IV at the time of capture, indicating that they were near their spawning grounds. Thus, it was inferred that their spawning grounds were located near the Hukou water area or within Poyang Lake. The exact locations of these spawning grounds require further research.

In this study, all *C. nasus* except for 6HK02 and 6HK05 theoretically originated from spawning grounds in the upper reaches of Poyang Lake or even Dongting Lake. However, *Fc* analyses across different waters indicated that most *C. nasus* continued to enter Poyang Lake for reproduction.

5. Conclusions

In this study, otolith microchemical techniques were used to investigate *C. nasus* in the Dawanzhou section of the Yangtze River. The results indicate that this section primarily serves as a migration channel for *C. nasus*. Additionally, the concept of the freshwater coefficient was introduced and it was speculated that most *C. nasus* in the Dawanzhou section may not continue migrating upstream but instead enter Poyang Lake through the Hukou water area for breeding and spawning. Therefore, the migration history of *C. nasus* in this section should be continuously monitored to supplement dynamic data on *C. nasus* in the lower reaches of the Yangtze River. This will help enhance conservation efforts and improve understanding of the ecological effects of the 10-year fishing ban in the Yangtze River.

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Abbreviations

The following abbreviations are used in this manuscript:

EPMA	X-ray electron probe microanalyzer
F_C	Freshwater coefficient

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